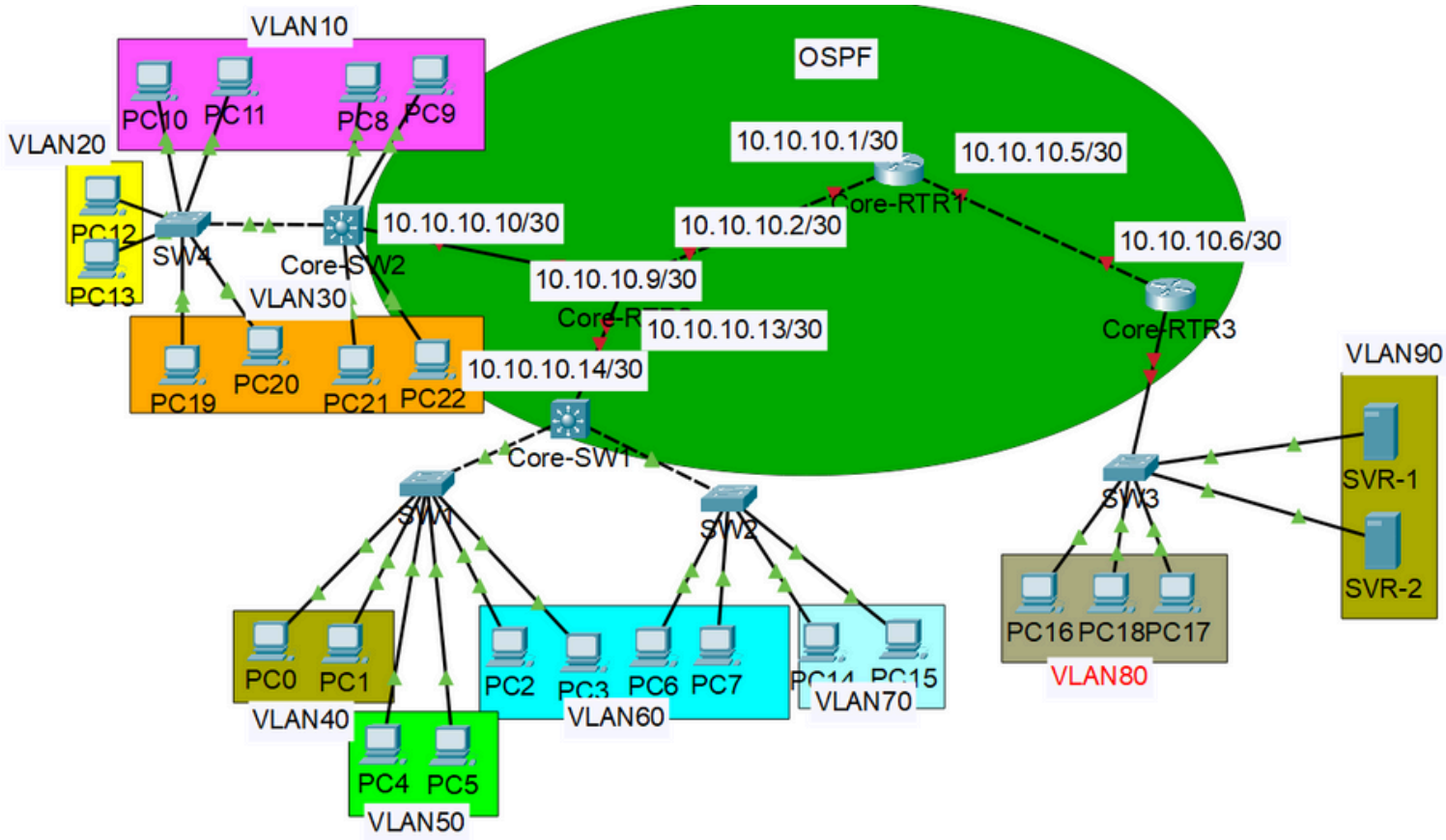
DESIGN BY FATHIN NAUFAL

NETWORK ENGINEER

# SUPERLAB CISCO CCNA 200-301

## SESSION 1




CISCO PACKET TRACER

# TOPICS SUPERLAB
# CISCO SESSION 1

# Topics

- VLAN & Trunking
- Portfast
- Router On Stick
- Switch Virtual Interface (SVI)
- DHCP Server
- Routing OSPF
- Access List
- DNS Server
- Web Server

# TASKS SUPERLAB
# CISCO SESSION 1

## Tasks:

### Part 1 - Basic Configuration

- Configure hostname in all of routers & switches same as display name
- Configure enable secret in all of routers & switches as "cisco"

### Part 2 - Configure VLAN & Trunking in Core-SW2 & SW4

- Create VLAN & set the vlan name as display
- Configure portfast in all of interfaces that connected to PC
- Configure SVI & DHCP Server in Core-SW2 for all VLAN, set the DNS Server to 192.168.90.2
- Bellow is the address detail of each VLAN. Core-SW2 will use first IP Address of each VLAN
- In the end of this section, make sure that each PC can ping each others

| VLAN | IP Address |
|---|---|
| VLAN10 | 192.168.10.0/24 |
| VLAN20 | 192.168.20.0/24 |
| VLAN30 | 192.168.30.0/24 |

### Part 3 - Configure VLAN & Trunking in Core-SW1, SW1, & SW2

- Create VLAN & set the vlan name as display
- Configure portfast in all of interfaces that connected to PC
- Configure SVI & DHCP Server in Core-SW1 for all VLAN, set the DNS Server to 192.168.90.2
- Bellow is the address detail of each VLAN. Core-SW1 will use first IP Address of each VLAN
- In the end of this section, make sure that each PC can ping each others

| VLAN | IP Address |
|---|---|
| VLAN40 | 192.168.40.0/24 |
| VLAN50 | 192.168.50.0/24 |
| VLAN60 | 192.168.60.0/24 |
| VLAN70 | 192.168.70.0/24 |

# TASKS SUPERLAB
# CISCO SESSION 1

## Part 4 - Configure VLAN & Trunking in Core-RTR3 & SW3

- Create VLAN & set the vlan name as display
- Configure portfast in all of interfaces that connected to PC
- Configure Router on Stick & DHCP Server in Core-RTR3 for VLAN80, set the DNS Server to 192.168.90.2
- Bellow is the address detail of each VLAN. Core-RTR3 will use first IP Address of each VLAN
- Configure IP Address on SVR1 & SVR2 manually, 192.168.90.2 for SVR1 and 192.168.90.3 for SVR2
- In the end of this section, make sure that each PC and Server can ping each others

| VLAN | IP Address |
|---|---|
| VLAN80 | 192.168.80.0/24 |
| VLAN90 | 192.168.90.0/24 |

## Part 5 - Configure Point to Point Addressing Between Routers

- Configure IP Address in all of routers as display
- In the end of this section, make sure that each routers can ping each others

## Part 6 - Configure OSPF in All of Routers

- Configure OSPF in all of routers as display
- In the end of this section, Make sure:
  - OSPF neighborship is UP in all of routers
  - Table routing is complete in all of routers
  - All of clients can ping each others

**Part 7 - DNS & Web Server Configuration**

The objective of this section is to set up DNS and Web services so that all clients can access the fathinnaufal.com site via a web browser.

**Tasks:**

1. **Configure the DNS Server on SVR1:**
   - Create an **A Record** for fathinnaufal.com that points to the IP address 192.168.90.3.
   - Create a **CNAME Record** for www.fathinnaufal.com as an alias for fathinnaufal.com.
2. **Configure the Web Server on SVR2:**
   - Set up the web service to display the message: **"Welcome to FathinNaufal.com"**.
3. **Verification:**
   - Ensure that all clients on the network can ping fathinnaufal.com.
   - Ensure that all clients can open fathinnaufal.com in a web browser.

## Part 8 - Configure Access List

- Configure Access List in Core-RTR3 to prevent access from VLAN10 & VLAN60 to Server
- Use access-list id 101
- In the end of this section, make sure that VLAN10 & VLAN60 can't access to server

# PART 1
# BASIC CONFIGURATION

## Part 1 - Basic Configuration

- Configure hostname in all of routers & switches same as display name
- Configure enable secret in all of routers & switches as "cisco"

**Rename & Enable Secret For Switch (SW1, SW2, SW3)**

```
Switch(config)#host SW1
SW1(config)#enable secret cisco

Switch(config)#hostname SW2
SW2(config)#enable secret cisco

Switch(config)#hostname SW3
SW3(config)#enable secret cisco

Switch(config)#host SW4
SW4(config)#enable secret cisco
```

**Rename & Enable Secret For Core Switch (Core-SW1, Core-SW2)**

```
Switch(config)#hostname Core-SW1
Core-SW1(config)#enable secret cisco

Switch(config)#host Core-SW2
Core-SW2(config)#enable secret cisco
```

**Rename & Enable Secret For Core Router (Core-RTR1, Core-RTR2, Core-RTR3)**

```
Router(config)#hostname Core-RTR1
Core-RTR1(config)#enable secret cisco

Router(config)#hostname Core-RTR2
Core-RTR2(config)#enable secret cisco

Router(config)#host Core-RTR3
Core-RTR3(config)#enable secret cisco
```

**DONT FORGET SAVE CONFIGURATION!**

```
Core-RTR1(config)#do write
Building configuration...
[OK]
```

# PART 2
# CONFIGURE VLAN & TRUNKING IN CORE-SW2 & SW4

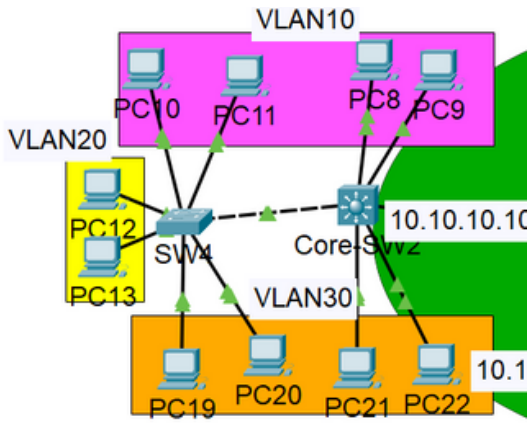
## Part 2 - Configure VLAN & Trunking in Core-SW2 & SW4

- Create VLAN & set the vlan name as display
- Configure portfast in all of interfaces that connected to PC
- Configure SVI & DHCP Server in Core-SW2 for all VLAN, set the DNS Server to 192.168.90.2
- Bellow is the address detail of each VLAN. Core-SW2 will use first IP Address of each VLAN
- In the end of this section, make sure that each PC can ping each others

### Configure VLAN SW 4, Access to VLAN, and Spanning-Tree Portfast and Trunk

```
SW4(config)#vlan 10
SW4(config-vlan)#name VLAN10
SW4(config-vlan)#vlan 20
SW4(config-vlan)#name VLAN20
SW4(config-vlan)#vlan 30
SW4(config-vlan)#name VLAN30
SW4(config-vlan)#int range fa0/2-3
SW4(config-if-range)#switchport mode access
SW4(config-if-range)#switchport acc vlan 10
SW4(config-if-range)#int range fa0/4-5
SW4(config-if-range)#switchport mode acc
SW4(config-if-range)#swit acc vl 20
SW4(config-if-range)#int range fa0/6-7
SW4(config-if-range)#switc mod acc
SW4(config-if-range)#swi acc vl 30
SW4(config-if-range)#int range fa0/2-7
SW4(config-if-range)#spanning-tree portfast

SW4(config)#int fa0/1
SW4(config-if)#switchport mode trunk
```




### Log Connected To Trunk

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1, changed state to up
```

### Check the configuration we have done! VLAN and Trunk (SW4)

```
SW4(config)#do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   VLAN10                           active    Fa0/2, Fa0/3
20   VLAN20                           active    Fa0/4, Fa0/5
30   VLAN30                           active    Fa0/6, Fa0/7

SW4(config)#do show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30
```

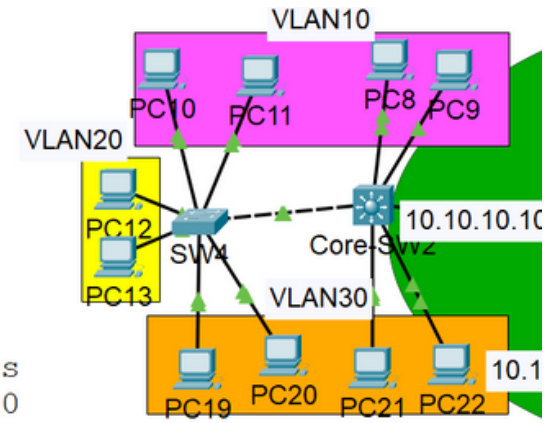
**DONT FORGET SAVE CONFIGURATION!**

```
SW4(config)#do write
Building configuration...
[OK]
```

# PART 2
# CONFIGURE VLAN & TRUNKING IN CORE-SW2 & SW4

## Configure VLAN Core-SW2, Access to VLAN. STP Portfast and Trunk

```
Core-SW2(config)#vlan 10
Core-SW2(config-vlan)#name VLAN10
Core-SW2(config-vlan)#vlan 20
Core-SW2(config-vlan)#name VLAN20
Core-SW2(config-vlan)#vlan 30
Core-SW2(config-vlan)#name VLAN30
```




```
Core-SW2(config-if-range)#switchport mode access
Core-SW2(config-if-range)#int range fa0/2-3
Core-SW2(config-if-range)#switchport mode access
Core-SW2(config-if-range)#switchport access vlan 10
Core-SW2(config)#int range fa0/4-5
Core-SW2(config-if-range)#switchport mode access
Core-SW2(config-if-range)#switchport acc vlan 30

Core-SW2(config-if-range)#int range fa0/2-5
Core-SW2(config-if-range)#spanning-tree portfast

Core-SW2(config)#int fa0/6
Core-SW2(config-if)#switchport trunk encapsulation dot1q
Core-SW2(config-if)#switchport mode trunk
```

## Check the configuration we have done! VLAN and Trunk (Core-SW2)

```
Core-SW2(config)#do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   VLAN10                           active    Fa0/2, Fa0/3
20   VLAN20                           active
30   VLAN30                           active    Fa0/4, Fa0/5


Core-SW2(config)#do show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/6       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/6       1-1005

Port        Vlans allowed and active in management domain
Fa0/6       1,10,20,30

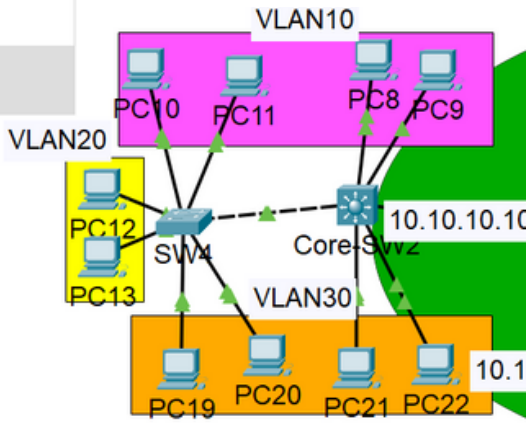
Port        Vlans in spanning tree forwarding state and not pruned
Fa0/6       1,10,20,30
```

**DONT FORGET SAVE CONFIGURATION!**

```
Core-SW2(config)#do write
Building configuration...
[OK]
```

# PART 2
# CONFIGURATION SVI & DHCP SERVER ON CORE-SW2

| VLAN | IP Address |
|---|---|
| VLAN10 | 192.168.10.0/24 |
| VLAN20 | 192.168.20.0/24 |
| VLAN30 | 192.168.30.0/24 |




## Configure IP on VLAN

```
Core-SW2(config)#int vlan 10
Core-SW2(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, cl

Core-SW2(config-if)#ip address 192.168.10.1 255.255.255.0
Core-SW2(config-if)#int vlan 20
Core-SW2(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

Core-SW2(config-if)#ip address 192.168.20.1 255.255.255.0
Core-SW2(config-if)#int vlan 30
Core-SW2(config-if)#
%LINK-5-CHANGED: Interface Vlan30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up

Core-SW2(config-if)#ip address 192.168.30.1 255.255.255.0
```

## Check the configuration we have done! SVI (Core-SW2)

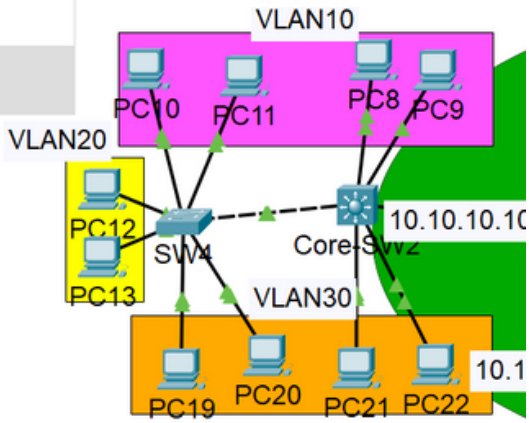
```
Core-SW2(config)#do show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.1    YES manual up                    up
Vlan20                 192.168.20.1    YES manual up                    up
Vlan30                 192.168.30.1    YES manual up                    up
```

**DONT FORGET SAVE CONFIGURATION!**

```
Core-SW2(config)#do write
Building configuration...
[OK]
```

# PART 2
# CONFIGURATION SVI & DHCP SERVER ON CORE-SW2

| VLAN | IP Address |
|---|---|
| VLAN10 | 192.168.10.0/24 |
| VLAN20 | 192.168.20.0/24 |
| VLAN30 | 192.168.30.0/24 |




## Configuration DHCP SERVER On Core-SW2

```
Core-SW2(config)#ip routing
Core-SW2(config)#ip dhcp pool VLAN10
Core-SW2(dhcp-config)#network 192.168.10.0 255.255.255
Core-SW2(dhcp-config)#default-router 192.168.10.1
Core-SW2(dhcp-config)#dns-server 192.168.90.2
Core-SW2(dhcp-config)#ip dhcp pool VLAN20
Core-SW2(dhcp-config)#network 192.168.20.0 255.255.255
Core-SW2(dhcp-config)#default-router 192.168.20.1
Core-SW2(dhcp-config)#dns-server 192.168.90.2
Core-SW2(config)#ip dhcp pool VLAN30
Core-SW2(dhcp-config)#network 192.168.30.0 255.255.255.0
Core-SW2(dhcp-config)#default-router 192.168.30.1
Core-SW2(dhcp-config)#dns-server 192.168.90.2
```

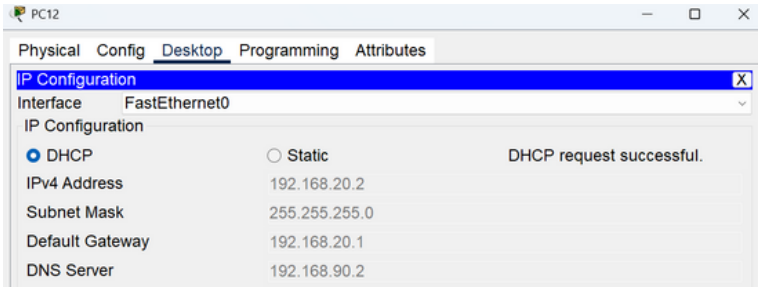
## Check the configuration we have done! DHCP-SERVER(Core-SW2)

```
Core-SW2(dhcp-config)#do show running-config | section dhcp
ip dhcp pool VLAN10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 192.168.90.2
ip dhcp pool VLAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 192.168.90.2
ip dhcp pool VLAN30
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 192.168.90.2
```

**DONT FORGET SAVE CONFIGURATION!**

```
Core-SW2(config)#do write
Building configuration...
[OK]
```

## Test DHCP-Server and Ping On PC




```
C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127

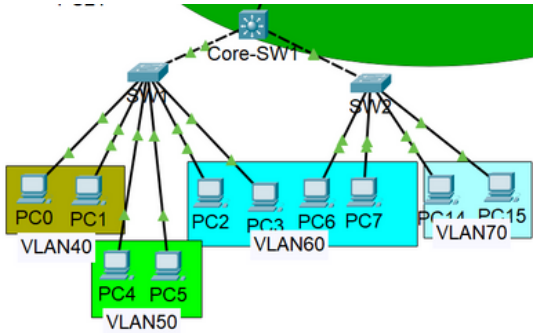
Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

# PART 3
# CONFIGURE VLAN & TRUNKING IN CORE-SW1, SW1, & SW2

## Part 3 - Configure VLAN & Trunking in Core-SW1, SW1, & SW2

- Create VLAN & set the vlan name as display
- Configure portfast in all of interfaces that connected to PC
- Configure SVI & DHCP Server in Core-SW1 for all VLAN, set the DNS Server to 192.168.90.2
- Bellow is the address detail of each VLAN. Core-SW1 will use first IP Address of each VLAN
- In the end of this section, make sure that each PC can ping each others

| VLAN | IP Address |
|---|---|
| VLAN40 | 192.168.40.0/24 |
| VLAN50 | 192.168.50.0/24 |
| VLAN60 | 192.168.60.0/24 |
| VLAN70 | 192.168.70.0/24 |




### Configure VLAN SW1, Access to VLAN, and Spanning-Tree Portfast and Trunk

```
SW1(config)#vlan 40
SW1(config-vlan)#name VLAN40
SW1(config-vlan)#vlan 50
SW1(config-vlan)#name VLAN50
SW1(config-vlan)#vlan 60
SW1(config-vlan)#name VLAN60
SW1(config-vlan)#int range fa0/2-3
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 40
SW1(config-if-range)#int range fa0/4-5
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 60
SW1(config-if-range)#int range fa0/6-7
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 50

SW1(config)#int range fa0/2-7
SW1(config-if-range)#spanning-tree portfast

SW1(config)#int fa0/1
SW1(config-if)#switchport mode trunk

SW1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1, changed state to up
```

**DONT FORGET SAVE CONFIGURATION!**

```
SW1(config)#do write
Building configuration...
[OK]
```

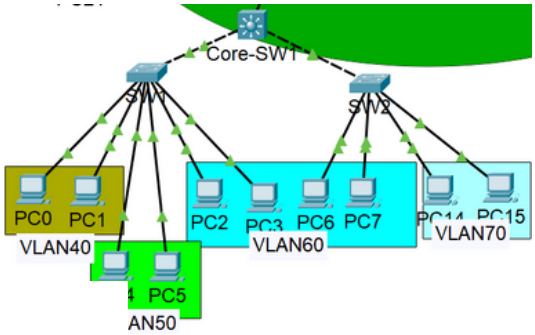
# PART 3
# CONFIGURE VLAN & TRUNKING ON CORE-SW1, SW1, & SW2

## Configure VLAN SW2, Access VLAN, and Spanning-Tree Portfast and Trunk

```
SW2(config)#vlan 60
SW2(config-vlan)#name VLAN60
SW2(config-vlan)#vlan 70
SW2(config-vlan)#name VLAN70
SW2(config-vlan)#int range fa0/2-3
SW2(config-if-range)#switchport mode access
SW2(config-if-range)#switchport access vlan 60
SW2(config-if-range)#int range fa0/4-5
SW2(config-if-range)#switchport mode access
SW2(config-if-range)#switchport access vlan 70
SW2(config-if-range)#int range fa0/2-5
SW2(config-if-range)#spanning-tree portfast
SW2(config)#int fa0/1
SW2(config-if)#switchport mode trunk

SW2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
```




## Result SW1, SW2 VLAN And Trunking

```
SW2(config)#do write
Building configuration...
[OK]
```

**DONT FORGET SAVE CONFIGURATION!**

```
SW1(config)#do show vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/8,
Fa0/9, Fa0/10, Fa0/11
                                                 Fa0/12,
Fa0/13, Fa0/14, Fa0/15
                                                 Fa0/16,
Fa0/17, Fa0/18, Fa0/19
                                                 Fa0/20,
Fa0/21, Fa0/22, Fa0/23
                                                 Fa0/24,
Gig0/1, Gig0/2
40   VLAN40                            active    Fa0/2, Fa0/3
50   VLAN50                            active    Fa0/6, Fa0/7
60   VLAN60                            active    Fa0/4, Fa0/5
```

```
SW2(config)#do show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/1,
Fa0/6, Fa0/7, Fa0/8
                                                 Fa0/9,
Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13,
Fa0/14, Fa0/15, Fa0/16
                                                 Fa0/17,
Fa0/18, Fa0/19, Fa0/20
                                                 Fa0/21,
Fa0/22, Fa0/23, Fa0/24
                                                 Gig0/1,
Gig0/2
60   VLAN60                            active    Fa0/2, Fa0/3
70   VLAN70                            active    Fa0/4, Fa0/5
```

```
SW1(config)#do show int trunk
Port        Mode         Encapsulation  Status        Native
vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,40,50,60

Port        Vlans in spanning tree forwarding state and not
pruned
Fa0/1       1,40,50,60
```

```
SW2(config)#do show int trunk
Port        Mode         Encapsulation  Status
Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

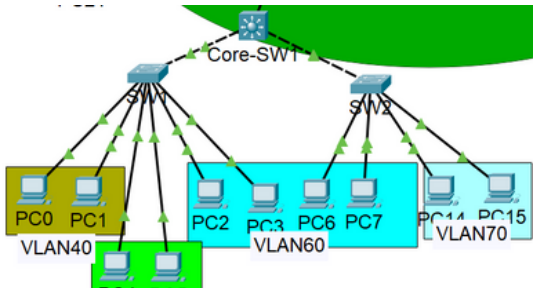
Port        Vlans allowed and active in management domain
Fa0/1       1,60,70

Port        Vlans in spanning tree forwarding state and
not pruned
Fa0/1       1,60,70
```

# PART 3
# CONFIGURE SVI & DHCP SERVER ON CORE-SW1

## Configure VLAN Core-SW1 and trunk dot1q

```
Core-SW1(config)#vlan 40
Core-SW1(config-vlan)#name VLAN40
Core-SW1(config-vlan)#vlan 50
Core-SW1(config-vlan)#name VLAN50
Core-SW1(config-vlan)#vlan 60
Core-SW1(config-vlan)#name VLAN60
Core-SW1(config-vlan)#vlan 70
Core-SW1(config-vlan)#name VLAN70
```




```
Core-SW1(config)#int range fa0/2-3
Core-SW1(config-if-range)#switchport trunk encapsulation dot1q
Core-SW1(config-if-range)#switchport mode trunk
```

## Configure IP on VLAN

**Dont Forget "ip routing"**

| VLAN | IP Address |
|---|---|
| VLAN40 | 192.168.40.0/24 |
| VLAN50 | 192.168.50.0/24 |
| VLAN60 | 192.168.60.0/24 |
| VLAN70 | 192.168.70.0/24 |

**DONT FORGET SAVE CONFIGURATION!**

```
Core-SW1(config)#do write
Building configuration...
[OK]
```

```
Core-SW1(config)#ip routing
Core-SW1(config)#int vlan 40
Core-SW1(config-if)#
%LINK-5-CHANGED: Interface Vlan40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up

Core-SW1(config-if)#ip address 192.168.40.1 255.255.255.0
Core-SW1(config-if)#int vlan 50
Core-SW1(config-if)#
%LINK-5-CHANGED: Interface Vlan50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan50, changed state to up

Core-SW1(config-if)#ip address 192.168.50.1 255.255.255.0

Core-SW1(config)#int vlan 60
Core-SW1(config-if)#
%LINK-5-CHANGED: Interface Vlan60, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan60, changed state to up

Core-SW1(config-if)#ip address 192.168.60.1 255.255.255.0
Core-SW1(config-if)#int vlan 70
Core-SW1(config-if)#
%LINK-5-CHANGED: Interface Vlan70, changed state to up

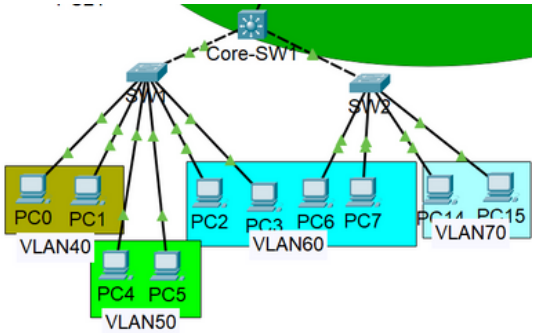
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan70, changed state to up

Core-SW1(config-if)#ip address 192.168.70.1 255.255.255.0
```

# PART 3
# CONFIGURE SVI & DHCP SERVER ON CORE-SW1

## Configure DHCP SERVER On Core-SW1

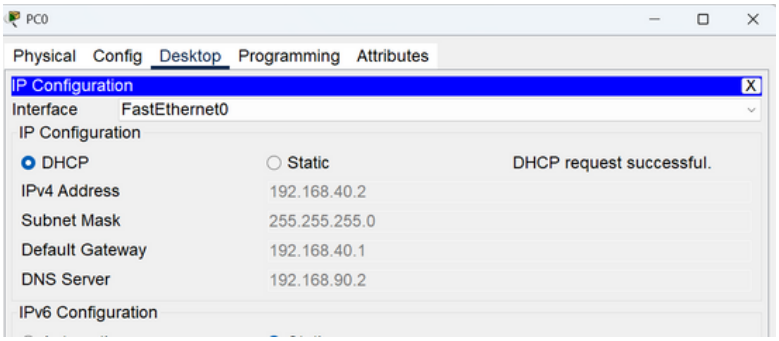
```
Core-SW1(config)#ip dhcp pool VLAN40
Core-SW1(dhcp-config)#network 192.168.40.0 255.255.255.0
Core-SW1(dhcp-config)#default-router 192.168.40.1
Core-SW1(dhcp-config)#dns-server 192.168.90.2
Core-SW1(dhcp-config)#ip dhcp pool VLAN50
Core-SW1(dhcp-config)#network 192.168.50.0 255.255.255.0
Core-SW1(dhcp-config)#default-router 192.168.50.1
Core-SW1(dhcp-config)#dns-server 192.168.90.2
Core-SW1(dhcp-config)#ip dhcp pool VLAN60
Core-SW1(dhcp-config)#network 192.168.60.0 255.255.255.0
Core-SW1(dhcp-config)#default-router 192.168.60.1
Core-SW1(dhcp-config)#dns-server 192.168.90.2
Core-SW1(dhcp-config)#ip dhcp pool VLAN70
Core-SW1(dhcp-config)#network 192.168.70.0 255.255.255.0
Core-SW1(dhcp-config)#default-router 192.168.70.1
Core-SW1(dhcp-config)#dns-server 192.168.90.2
```

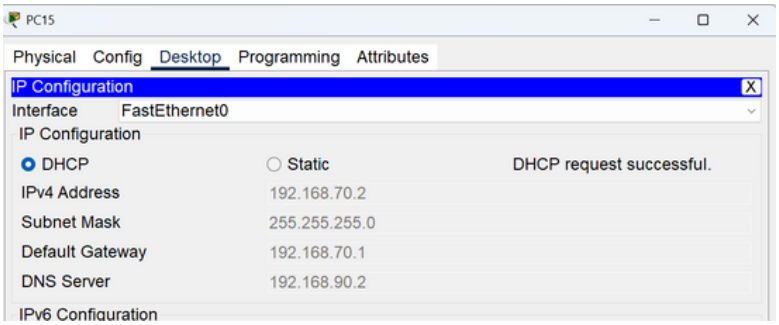



**DONT FORGET SAVE CONFIGURATION!**

```
Core-SW1(config)#do write
Building configuration...
[OK]
```

## Result DHCP SERVER to User




## Test Ping PC VLAN40 to PC VLAN70

```
C:\>ping 192.168.70.2

Pinging 192.168.70.2 with 32 bytes of data:

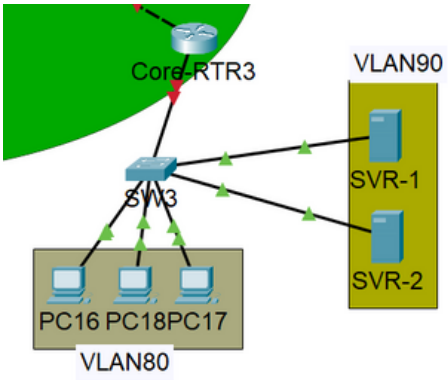
Reply from 192.168.70.2: bytes=32 time<1ms TTL=127
Reply from 192.168.70.2: bytes=32 time<1ms TTL=127
Reply from 192.168.70.2: bytes=32 time<1ms TTL=127
Reply from 192.168.70.2: bytes=32 time<1ms TTL=127
```

# PART 4
# VLAN & TRUNKING
# IN SW3 & CORE-RTR3

## Part 4 - Configure VLAN & Trunking in Core-RTR3 & SW3

- Create VLAN & set the vlan name as display
- Configure portfast in all of interfaces that connected to PC
- Configure Router on Stick & DHCP Server in Core-RTR3 for VLAN80, set the DNS Server to 192.168.90.2
- Bellow is the address detail of each VLAN. Core-RTR3 will use first IP Address of each VLAN
- Configure IP Address on SVR1 & SVR2 manually, 192.168.90.2 for SVR1 and 192.168.90.3 for SVR2
- In the end of this section, make sure that each PC and Server can ping each others

### Configure VLAN and Access VLAN, Trunk, STP Portfast SW3




```
SW3(config)#vlan 80
SW3(config-vlan)#name VLAN80
SW3(config-vlan)#vlan 90
SW3(config-vlan)#name VLAN90

SW3(config)#int range fa0/4-6
SW3(config-if-range)#switchport mode access
SW3(config-if-range)#switchport access vlan 80
SW3(config-if-range)#int range fa0/2-3
SW3(config-if-range)#switchport mode access
SW3(config-if-range)#switchport access vlan 90

SW3(config-if)#int range fa0/2-6
SW3(config-if-range)#spanning-tree portfast

SW3(config)#int fa0/1
SW3(config-if)#switchport mode trunk
SW3(config-if)#do show int trunk


SW3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed stat
to up
```

**DONT FORGET SAVE CONFIGURATION!**

```
SW3(config-if)#do write
Building configuration...
[OK]
```

### Result Configure SW3

```
SW3(config)#do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
80   VLAN80                           active    Fa0/4, Fa0/5, Fa0/6
90   VLAN90                           active    Fa0/2, Fa0/3
```

# PART 4
# VLAN & TRUNKING
# IN SW3 & CORE-RTR3

| VLAN | IP Address |
|---|---|
| VLAN80 | 192.168.80.0/24 |
| VLAN90 | 192.168.90.0/24 |

## Configure IP Address And DHCP Server Core-RTR3

```
Core-RTR3(config)#int fa0/1.80
Core-RTR3(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.80, changed state to up

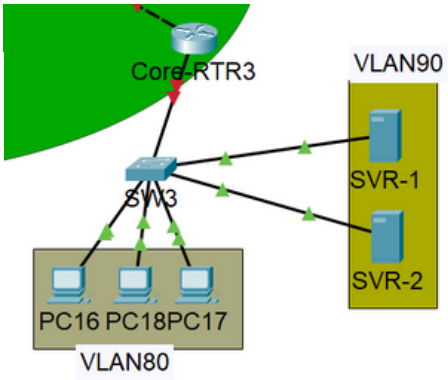
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.80, changed state to up

Core-RTR3(config-subif)#encapsulation dot1q 80
Core-RTR3(config-subif)#ip address 192.168.80.1 255.255.255.0
Core-RTR3(config-subif)#int fa0/1.90
Core-RTR3(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.90, changed state to up

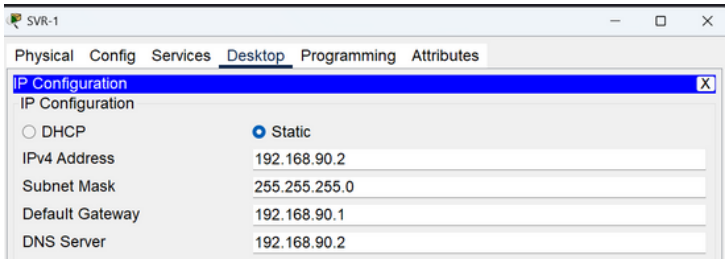
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.90, changed state to up

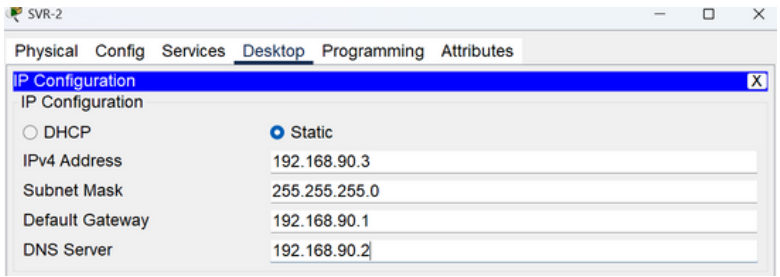
Core-RTR3(config-subif)#encapsulation dot1q 90
Core-RTR3(config-subif)#ip address 192.168.90.1 255.255.255.0

Core-RTR3(config)#ip dhcp pool VLAN80
Core-RTR3(dhcp-config)#network 192.168.80.0 255.255.255.0
Core-RTR3(dhcp-config)#default-router 192.168.80.1
Core-RTR3(dhcp-config)#dns-server 192.168.90.2
```




## Configure SVR-1 and SVR-2 Manual




## Test Ping PC To Server

```
C:\>ping 192.168.90.2

Pinging 192.168.90.2 with 32 bytes of data:

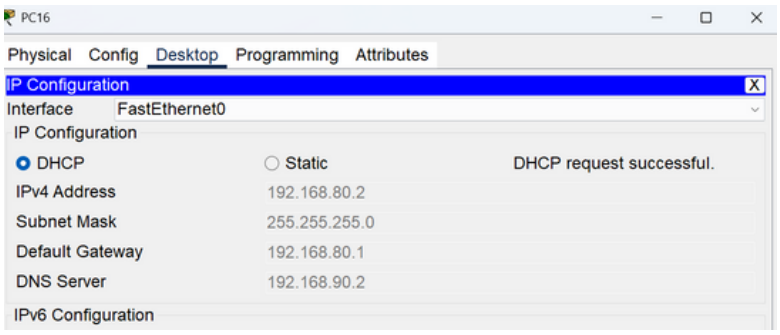
Reply from 192.168.90.2: bytes=32 time<1ms TTL=127
Reply from 192.168.90.2: bytes=32 time<1ms TTL=127
Reply from 192.168.90.2: bytes=32 time<1ms TTL=127
Reply from 192.168.90.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.90.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

**DONT FORGET SAVE CONFIGURATION!**

```
Core-RTR3(config)#do write
Building configuration...
[OK]
```

## Result Configure DHCP SERVER



# PART 5
# CONFIGURE POINT TO POINT ADDRESSING BETWEEN ROUTERS

## Part 5 - Configure Point to Point Addressing Between Routers

- Configure IP Address in all of routers as display
- In the end of this section, make sure that each routers can ping each others

### Configure Ip Address On Routers (Core-RTR1)

**DONT FORGET SAVE CONFIGURATION!**

```
Core-RTR1(config)#int range fa0/0-1
Core-RTR1(config-if-range)#no shutdown

Core-RTR1(config-if-range)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Core-RTR1(config-if-range)#exit
Core-RTR1(config)#int fa0/1
Core-RTR1(config-if)#ip address 10.10.10.5 255.255.255.252
Core-RTR1(config-if)#int fa0/0
Core-RTR1(config-if)#ip address 10.10.10.1 255.255.255.252

Core-RTR1(config-if)#do show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.10.1      YES manual up                    up
FastEthernet0/1        10.10.10.5      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
Core-RTR1(config)#do write
Building configuration...
[OK]
```

### Configure Ip Address On Routers (Core-RTR2)

**DONT FORGET SAVE CONFIGURATION!**

```
Core-RTR2(config-if-range)#int range fa0/0-1, fa1/0
Core-RTR2(config-if-range)#no shutdown

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Core-RTR2(config)#int fa1/0
Core-RTR2(config-if)#ip address 10.10.10.2 255.255.255.252
Core-RTR2(config-if)#int fa0/1
Core-RTR2(config-if)#ip address 10.10.10.9 255.255.255.252

Core-RTR2(config)#int fa0/0
Core-RTR2(config-if)#ip address 10.10.10.13 255.255.255.252
Core-RTR2(config-if)#do show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.10.13     YES manual up                    up
FastEthernet0/1        10.10.10.9      YES manual up                    up
FastEthernet1/0        10.10.10.2      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
Core-RTR2(config)#do write
Building configuration...
[OK]
```

### Configure Ip Address On Routers (Core-RTR3)

**DONT FORGET SAVE CONFIGURATION!**

```
Core-RTR3(config)#int fa0/0
Core-RTR3(config-if)#no shutdown

Core-RTR3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Core-RTR3(config-if)#ip address 10.10.10.6 255.255.255.252

Core-RTR3(config)#do show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.10.6      YES manual up                    down
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/1.80     192.168.80.1    YES manual up                    up
FastEthernet0/1.90     192.168.90.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
Core-RTR3(config)#do write
Building configuration...
[OK]
```

# PART 5 CONFIGURE POINT TO POINT ADDRESSING BETWEEN CORE SWITCH

## Configure Ip Address On Core Switch (Core-SW1)

```
Core-SW1(config)#int fa0/1
Core-SW1(config-if)#no switchport
Core-SW1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1, changed state to up
Core-SW1(config-if)#ip address 10.10.10.14 255.255.255.252
Core-SW1(config)#do show ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        10.10.10.14     YES manual up                    up
```

## Configure Ip Address On Core Switch (Core-SW2)

```
Core-SW2(config)#int fa0/1
Core-SW2(config-if)#no switchport
Core-SW2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Core-SW2(config-if)#ip address 10.10.10.10 255.255.255.252
Core-SW2(config-if)#do show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        10.10.10.10     YES manual up                    up
```

**DONT FORGET SAVE CONFIGURATION!**

```
Core-SW1(config)#do write
Building configuration...
[OK]
```

```
Core-SW2(config)#do write
Building configuration...
[OK]
```

# PART 5
# TEST PING CORE ROUTER AND CORE SWITCH

## Test Ping Core-RTR1 to Core-RTR3

```
Core-RTR1(config)#do ping 10.10.10.6

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.6, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

## Test Ping Core-RTR1 to Core-RTR2

```
Core-RTR1(config)#do ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

## Test Ping Core-RTR2 to Core-RTR1

```
Core-RTR2(config)#do ping 10.10.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

## Test Ping Core-RTR2 to Core-SW1

```
Core-RTR2(config)#do ping 10.10.10.14

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.14, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/8 ms
```

## Test Ping Core-RTR2 to Core-SW2

```
Core-RTR2(config)#do ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

## Test Ping Core-SW1 to Core-RTR2

```
Core-SW1(config)#do ping 10.10.10.13

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.13, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

## Test Ping Core-SW2 to Core-RTR2

```
Core-SW2(config)#do ping 10.10.10.9

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.9, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

# PART 6
# CONFIGURE OSPF IN ALL OF ROUTERS AND CORE SWITCH

## Part 6 - Configure OSPF in All of Routers

- Configure OSPF in all of routers as display
- In the end of this section, Make sure:
  - OSPF neighborship is UP in all of routers
  - Table routing is complete in all of routers
  - All of clients can ping each others

**DONT FORGET SAVE CONFIGURATION!**

```
Core-RTR1(config)#do write
Building configuration...
[OK]
```

### Routing OSPF In Core-RTR1

```
Core-RTR1(config)#router ospf 1
Core-RTR1(config-router)#network 10.10.10.4 0.0.0.3 area 0
Core-RTR1(config-router)#network 10.10.10.0 0.0.0.3 area 0
```

**DONT FORGET SAVE CONFIGURATION!**

```
Core-RTR2(config)#do write
Building configuration...
[OK]
```

### Routing OSPF In Core-RTR2

```
Core-RTR2(config)#router ospf 1
Core-RTR2(config-router)#network 10.10.10.12 0.0.0.3 area 0
Core-RTR2(config-router)#network 10.10.10.0 0.0.0.3 area 0
Core-RTR2(config-router)#
00:08:56: %OSPF-5-ADJCHG: Process 1, Nbr 10.10.10.5 on FastEthernet1/0
from LOADING to FULL, Loading Done

Core-RTR2(config-router)#network 10.10.10.8 0.0.0.3 area 0
```

**DONT FORGET SAVE CONFIGURATION!**

```
Core-RTR3(config)#do write
Building configuration...
[OK]
```

### Routing OSPF In Core-RTR3

```
Core-RTR3(config)#router ospf 1
Core-RTR3(config-router)#network 10.10.10.4 0.0.0.3 area 0
Core-RTR3(config-router)#
00:05:08: %OSPF-5-ADJCHG: Process 1, Nbr 10.10.10.5 on
FastEthernet0/0 from LOADING to FULL, Loading Done

Core-RTR3(config-router)#network 192.168.80.0 0.0.0.255 area 0
Core-RTR3(config-router)#network 192.168.90.0 0.0.0.255 area 0
```

**DONT FORGET SAVE CONFIGURATION!**

```
Core-SW1(config)#do write
Building configuration...
[OK]
```

### Routing OSPF In Core-SW1

```
Core-SW1(config)#router ospf 1
Core-SW1(config-router)#network 10.10.10.12 0.0.0.3 area 0
Core-SW1(config-router)#
16:12:57: %OSPF-5-ADJCHG: Process 1, Nbr 10.10.10.13 on FastEthernet0/1 from LOADING
to FULL, Loading Done

Core-SW1(config-router)#network 192.168.40.0 0.0.0.255 area 0
Core-SW1(config-router)#network 192.168.50.0 0.0.0.255 area 0
Core-SW1(config-router)#network 192.168.60.0 0.0.0.255 area 0
Core-SW1(config-router)#network 192.168.70.0 0.0.0.255 area 0
```

**DONT FORGET SAVE CONFIGURATION!**

```
Core-SW2(config)#do write
Building configuration...
[OK]
```

### Routing OSPF In Core-SW2

```
Core-SW2(config)#router ospf 1
Core-SW2(config-router)#network 10.10.10.8 0.0.0.3 area 0
Core-SW2(config-router)#network 192.168.10.0 0.0.0.255 area 0
Core-SW2(config-router)#network 192.168.20.0 0.0.0.255 area 0
Core-SW2(config-router)#network 192.168.30.0 0.0.0.255 area 0
```

# PART 6 CHECK TABLE ROUTE OSPF IN ALL OF ROUTERS

## Table Routing OSPF In Core-RTR1

```
     10.0.0.0/30 is subnetted, 4 subnets
C       10.10.10.0 is directly connected, FastEthernet0/0
C       10.10.10.4 is directly connected, FastEthernet0/1
O       10.10.10.8 [110/2] via 10.10.10.2, 00:09:28, FastEthernet0/0
O       10.10.10.12 [110/2] via 10.10.10.2, 00:18:12, FastEthernet0/0
O    192.168.10.0/24 [110/3] via 10.10.10.2, 00:09:18, FastEthernet0/0
O    192.168.20.0/24 [110/3] via 10.10.10.2, 00:09:18, FastEthernet0/0
O    192.168.30.0/24 [110/3] via 10.10.10.2, 00:09:18, FastEthernet0/0
O    192.168.40.0/24 [110/3] via 10.10.10.2, 00:06:43, FastEthernet0/0
O    192.168.50.0/24 [110/3] via 10.10.10.2, 00:06:28, FastEthernet0/0
O    192.168.60.0/24 [110/3] via 10.10.10.2, 00:06:18, FastEthernet0/0
O    192.168.70.0/24 [110/3] via 10.10.10.2, 00:06:18, FastEthernet0/0
O    192.168.80.0/24 [110/2] via 10.10.10.6, 00:21:30, FastEthernet0/1
O    192.168.90.0/24 [110/2] via 10.10.10.6, 00:21:20, FastEthernet0/1
```

## Table Routing OSPF In Core-RTR2

```
     10.0.0.0/30 is subnetted, 4 subnets
C       10.10.10.0 is directly connected, FastEthernet1/0
O       10.10.10.4 [110/2] via 10.10.10.1, 00:18:50, FastEthernet1/0
C       10.10.10.8 is directly connected, FastEthernet0/1
C       10.10.10.12 is directly connected, FastEthernet0/0
O    192.168.10.0/24 [110/2] via 10.10.10.10, 00:10:06, FastEthernet0/1
O    192.168.20.0/24 [110/2] via 10.10.10.10, 00:10:06, FastEthernet0/1
O    192.168.30.0/24 [110/2] via 10.10.10.10, 00:10:06, FastEthernet0/1
O    192.168.40.0/24 [110/2] via 10.10.10.14, 00:07:21, FastEthernet0/0
O    192.168.50.0/24 [110/2] via 10.10.10.14, 00:07:05, FastEthernet0/0
O    192.168.60.0/24 [110/2] via 10.10.10.14, 00:06:55, FastEthernet0/0
O    192.168.70.0/24 [110/2] via 10.10.10.14, 00:06:55, FastEthernet0/0
O    192.168.80.0/24 [110/3] via 10.10.10.1, 00:18:50, FastEthernet1/0
O    192.168.90.0/24 [110/3] via 10.10.10.1, 00:18:50, FastEthernet1/0
```

## Table Routing OSPF In Core-RTR3

```
     10.0.0.0/30 is subnetted, 4 subnets
O       10.10.10.0 [110/2] via 10.10.10.5, 00:19:49, FastEthernet0/0
C       10.10.10.4 is directly connected, FastEthernet0/0
O       10.10.10.8 [110/3] via 10.10.10.5, 00:11:05, FastEthernet0/0
O       10.10.10.12 [110/3] via 10.10.10.5, 00:19:49, FastEthernet0/0
O    192.168.10.0/24 [110/4] via 10.10.10.5, 00:11:05, FastEthernet0/0
O    192.168.20.0/24 [110/4] via 10.10.10.5, 00:11:05, FastEthernet0/0
O    192.168.30.0/24 [110/4] via 10.10.10.5, 00:11:05, FastEthernet0/0
O    192.168.40.0/24 [110/4] via 10.10.10.5, 00:08:25, FastEthernet0/0
O    192.168.50.0/24 [110/4] via 10.10.10.5, 00:08:09, FastEthernet0/0
O    192.168.60.0/24 [110/4] via 10.10.10.5, 00:07:59, FastEthernet0/0
O    192.168.70.0/24 [110/4] via 10.10.10.5, 00:07:59, FastEthernet0/0
C    192.168.80.0/24 is directly connected, FastEthernet0/1.80
C    192.168.90.0/24 is directly connected, FastEthernet0/1.90
```

## Test Ping Core-RTR2 to PC VLAN80

```
Core-RTR2(config)#do ping 192.168.80.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.80.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Core-RTR2(config)#do traceroute 192.168.80.2
Type escape sequence to abort.
Tracing the route to 192.168.80.2

  1   10.10.10.1      0 msec    0 msec    0 msec
  2   10.10.10.6      0 msec    0 msec    0 msec
  3   192.168.80.2    0 msec    0 msec    5 msec
```

# PART 6 CHECK TABLE ROUTE OSPF IN ALL OF CORE-SWITCH

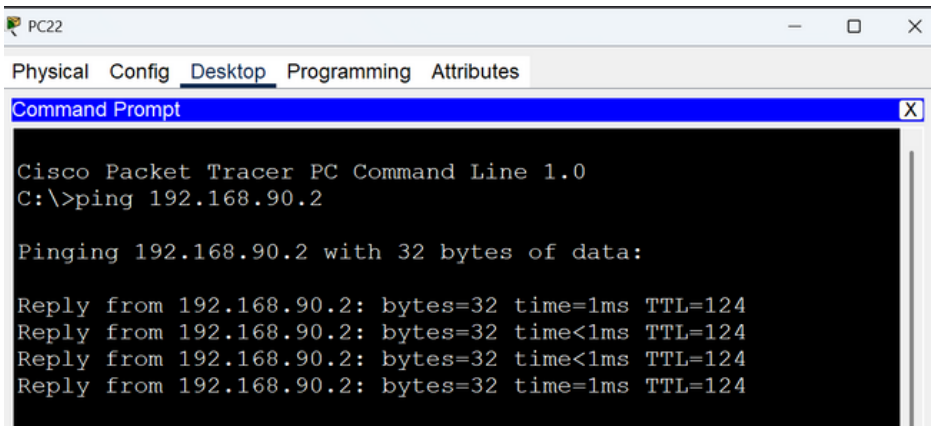
## Table Routing OSPF In Core-SW1

```
      10.0.0.0/30 is subnetted, 4 subnets
O        10.10.10.0 [110/2] via 10.10.10.13, 00:20:03, FastEthernet0/1
O        10.10.10.4 [110/3] via 10.10.10.13, 00:20:03, FastEthernet0/1
O        10.10.10.8 [110/2] via 10.10.10.13, 00:16:22, FastEthernet0/1
C        10.10.10.12 is directly connected, FastEthernet0/1
O     192.168.10.0/24 [110/3] via 10.10.10.13, 00:16:22, FastEthernet0/1
O     192.168.20.0/24 [110/3] via 10.10.10.13, 00:16:22, FastEthernet0/1
O     192.168.30.0/24 [110/3] via 10.10.10.13, 00:16:22, FastEthernet0/1
C     192.168.40.0/24 is directly connected, Vlan40
C     192.168.50.0/24 is directly connected, Vlan50
C     192.168.60.0/24 is directly connected, Vlan60
C     192.168.70.0/24 is directly connected, Vlan70
O     192.168.80.0/24 [110/4] via 10.10.10.13, 00:20:03, FastEthernet0/1
O     192.168.90.0/24 [110/4] via 10.10.10.13, 00:20:03, FastEthernet0/1
```

## Table Routing OSPF In Core-SW2

```
      10.0.0.0/30 is subnetted, 4 subnets
O        10.10.10.0 [110/2] via 10.10.10.9, 00:16:53, FastEthernet0/1
O        10.10.10.4 [110/3] via 10.10.10.9, 00:16:53, FastEthernet0/1
C        10.10.10.8 is directly connected, FastEthernet0/1
O        10.10.10.12 [110/2] via 10.10.10.9, 00:16:53, FastEthernet0/1
C     192.168.10.0/24 is directly connected, Vlan10
C     192.168.20.0/24 is directly connected, Vlan20
C     192.168.30.0/24 is directly connected, Vlan30
O     192.168.40.0/24 [110/3] via 10.10.10.9, 00:14:11, FastEthernet0/1
O     192.168.50.0/24 [110/3] via 10.10.10.9, 00:13:55, FastEthernet0/1
O     192.168.60.0/24 [110/3] via 10.10.10.9, 00:13:45, FastEthernet0/1
O     192.168.70.0/24 [110/3] via 10.10.10.9, 00:13:45, FastEthernet0/1
O     192.168.80.0/24 [110/4] via 10.10.10.9, 00:16:53, FastEthernet0/1
O     192.168.90.0/24 [110/4] via 10.10.10.9, 00:16:53, FastEthernet0/1
```

## Test Ping PC22 (VLAN 30) To Server




```
C:\>tracert 192.168.90.2

Tracing route to 192.168.90.2 over a maximum of 30 hops:

  1    0 ms      1 ms      0 ms      192.168.30.1
  2    0 ms      0 ms      0 ms      10.10.10.9
  3    0 ms      0 ms      0 ms      10.10.10.1
  4    0 ms      1 ms      0 ms      10.10.10.6
  5    1 ms      0 ms      0 ms      192.168.90.2

Trace complete.
```

# PART 7
# CONFIGURE DNS & WEB SERVER

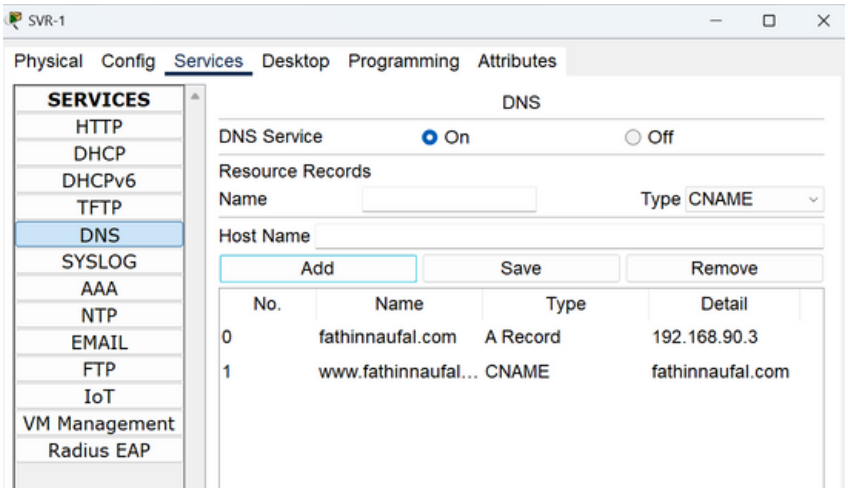
**Part 7 - DNS & Web Server Configuration**

The objective of this section is to set up DNS and Web services so that all clients can access the fathinnaufal.com site via a web browser.

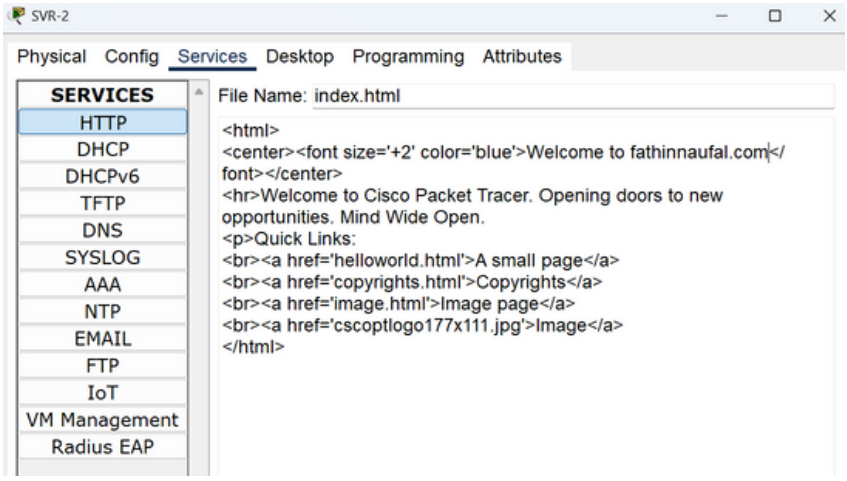
**Tasks:**

1. **Configure the DNS Server on SVR1:**
    - Create an **A Record** for fathinnaufal.com that points to the IP address 192.168.90.3.
    - Create a **CNAME Record** for www.fathinnaufal.com as an alias for fathinnaufal.com.
2. **Configure the Web Server on SVR2:**
    - Set up the web service to display the message: **"Welcome to FathinNaufal.com"**.
3. **Verification:**
    - Ensure that all clients on the network can ping fathinnaufal.com.
    - Ensure that all clients can open fathinnaufal.com in a web browser.

## Configure DNS In SVR-1




## Configure Web Server In SVR-2



# PART 7
# TEST DNS & WEB SERVER

## Test Ping PC22 (VLAN30) to DNS Server

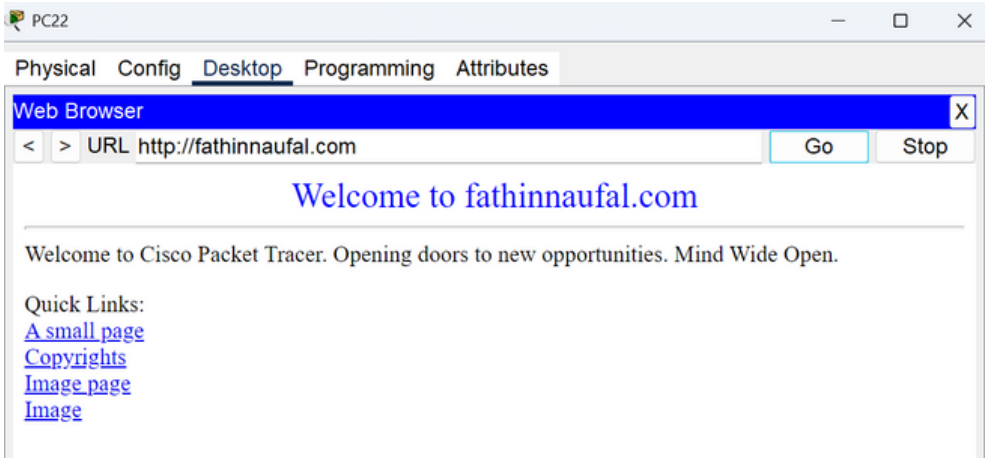
```
C:\>ping fathinnaufal.com

Pinging 192.168.90.3 with 32 bytes of data:

Reply from 192.168.90.3: bytes=32 time<1ms TTL=124
Reply from 192.168.90.3: bytes=32 time=1ms TTL=124
Reply from 192.168.90.3: bytes=32 time<1ms TTL=124
Reply from 192.168.90.3: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.90.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

## Test Ping PC22 (VLAN30) to Web Server



# PART 8
# CONFIGURE ACCESS LIST

## Part 8 - Configure Access List

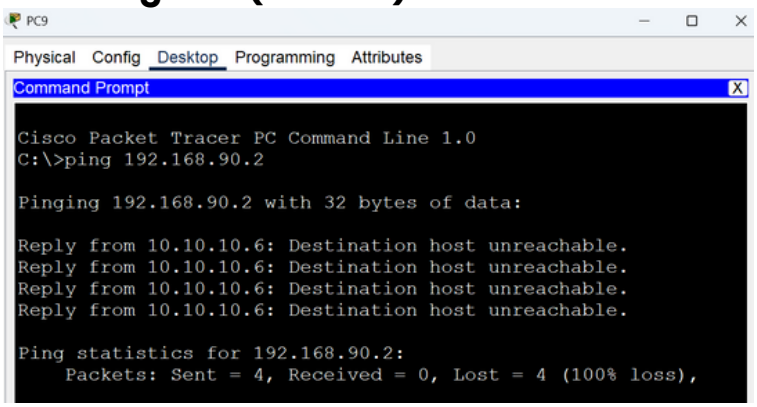
- Configure Access List in Core-RTR3 to prevent access from VLAN10 & VLAN60 to Server
- Use access-list id 101
- In the end of this section, make sure that VLAN10 & VLAN60 can't access to server

**Case :** Core-RTR3 to prevent access from VLAN10 & VLAN60 to Server

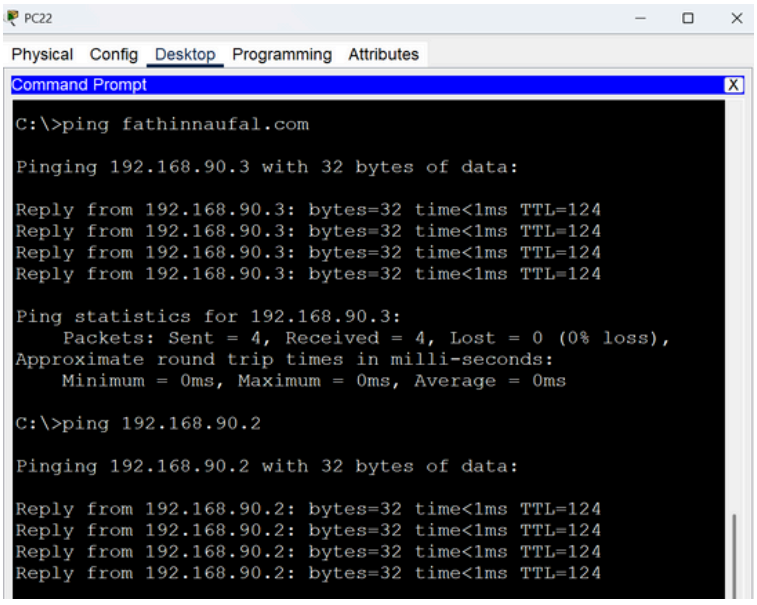
```
Core-RTR3(config)#access-list 101 deny ip 192.168.10.0 0.0.0.255 192.168.90.0 0.0.0.255
Core-RTR3(config)#access-list 101 deny ip 192.168.60.0 0.0.0.255 192.168.90.0 0.0.0.255
Core-RTR3(config)#access-list 101 permit ip any any
Core-RTR3(config)#int fa0/1.90
Core-RTR3(config-subif)#ip access-group 101 out

Core-RTR3(config)#do show access-list
Extended IP access list 101
    10 deny ip 192.168.10.0 0.0.0.255 192.168.90.0 0.0.0.255
    20 deny ip 192.168.60.0 0.0.0.255 192.168.90.0 0.0.0.255
    30 permit ip any any
```

### Test Ping PC9 (VLAN10) to Server Blocked with Access List




### Test Ping PC22 (VLAN30) to Server



# SUMMARY PART 1 - PART 8

**Superlab Configuration Summary**

**Part 1: Basic Device Configuration**

- Assigning hostnames according to the topology and setting the enable secret password to "cisco" on all routers and switches in the network.

**Part 2: VLAN & DHCP on Core-SW2**

- Creating VLANs 10, 20, and 30. Then, configuring Switch Virtual Interfaces (SVIs) on Core-SW2 to act as a gateway and as a DHCP server for these three VLANs.

**Part 3: VLAN & DHCP on Core-SW1**

- Performing a similar configuration to Part 2, but for VLANs 40, 50, 60, and 70. Core-SW1 is set up as the gateway (SVI) and DHCP server for these VLANs.

**Part 4: Router on a Stick & Servers**

- Configuring Core-RTR3 as a gateway for VLANs 80 and 90 using the "Router on a Stick" method. This router also acts as a DHCP server for VLAN 80 , while the servers in VLAN 90 (SVR-1 and SVR-2) are assigned static IP addresses.

**Part 5: Point-to-Point Addressing Between Core Devices**

- Assigning IP addresses to the interfaces connecting all core devices (routers and switches) to establish point-to-point connectivity between them.

**Part 6: OSPF Dynamic Routing**

- Activating the OSPF routing protocol on all core devices to automatically exchange route information , allowing all devices across all VLANs to communicate with each other.

**Part 7: DNS & Web Server Configuration**

- Setting up SVR-1 as a DNS server to resolve the domain name fathinnaufal.com. SVR-2 is configured as a web server to host the site for that domain.

**Part 8: Security with Access Control List (ACL)**

- Creating a security rule (Access List 101) on Core-RTR3 to block access from VLAN 10 and VLAN 60 to the server network (VLAN 90)